

Project Title :

AZURE LOAD BALANCER OUTBOUND RULES AND SNAT PLANNING

Project Number : P072

Problem Statement :

Generally application might make thousands of requests and every outbound connection needs a source port. So Azure Load balancers uses SNAT(Source Network Address Translation) for communication between the virtual machine with internet using public IP address of the Load balancer. Each virtual machine makes lot of outgoing connections. But each VM has limited number of SNAT ports available. If all the available ports are being used it may result in connection failure or delays. This is known as SNAT port exhaustion. This cause outbound connection timeouts and application performance issues

USE CASES:

1. Prevent SNAT port exhaustion for a chatty application

A chatty microservice(that talks to other services far more often than it needs) can burn through SNAT ports faster than they're released, causing intermittent connection failures that look like application bugs rather than a capacity issue. Learn to confirm the real cause via SNAT metrics and fix it with outbound rules, more frontend IPs, or reduced connection churn.

2. Size outbound capacity correctly before production

Default SNAT port allocation shrinks per instance as the backend pool scales out, so headroom at low scale can become exhaustion at peak. Practising the sizing formulas and configure frontend IPs and outbound rules to hold up under real autoscale conditions, not just current scale

Objectives

- To understand the purpose of SNAT and how it is useful for the communication between Virtual machines with the internet
- To identify the causes of SNAT port exhaustion in applications during multiple outbound connections.
- To configure Azure Load Balancer outbound rules to prevent SNAT port exhaustion.
- To calculate the assigning of SNAT ports to the virtual machines for port allocation.

Requirements

- An active Azure subscription with permissions to create Load Balancers, Virtual Machines/VMSS, Virtual networks and Network Security Groups
- Standard SKU Load Balancer (Basic SKU does not support outbound rules)
- A backend pool of VMs or a VM Scale Set that can run a simple load-generation script
- Azure Monitor enabled with diagnostic settings on the Load Balancer to capture SNAT metrics (Used SNAT Ports, Allocated SNAT Ports, SNAT Connection Count)
- A shared external endpoint to send outbound test traffic to (e.g., a public HTTP echo/test service)
- Ability to scale the backend pool up/down during the exercise (manually or via auto scale rules)

Abstract

Cloud applications interact with many databases, external APIs, and many other cloud services. While interacting with these many services, they create many outbound connections. For maintaining the outbound traffic, cloud applications usually have load balancers, which send the outbound traffic through the Source Network Address Translation (SNAT) ports. But if the outbound connections increase too heavily, the SNAT ports in the Azure application may become insufficient and may cause many application level problems.

The project mainly focuses on Azure Load Balancer, its outbound rules and how the SNAT ports are being planned to control the outbound traffic. For example, if we take any chatty application like an online shopping website, we have to study and analyse the outbound traffic and allocate the SNAT ports for different connections. We will design this application using Azure Load balancer, virtual machines and the required outbound connectivity. Then for testing purpose, we will send the sample outbound traffic through these SNAT ports and these SNAT ports have to analyse the outbound rules and we have to observe how well the SNAT ports are being able to control this traffic. Observe the SNAT port exhaustion and the traffic that each SNAT port is being able to handle.

In this project, it should observe and estimate the outbound traffic capacity for the SNAT ports, so that the port exhaustion can be reduced. For this, the SNAT ports will be tested by sending different configurations of outbound traffic through them. After analysing these configurations of traffic, the SNAT ports will be able to estimate the traffic and should maintain the ports such that the port exhaustion should be reduced.

At the end, this project's outcome should provide a practical understanding of Azure Load Balancer, outbound rules and how the SNAT ports are going to handle these. And also it shows the proper SNAT planning can improve the scalability of the cloud applications.

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